

Demonstrating Assembler³: 3D Reconstruction of Laser-Cut Models

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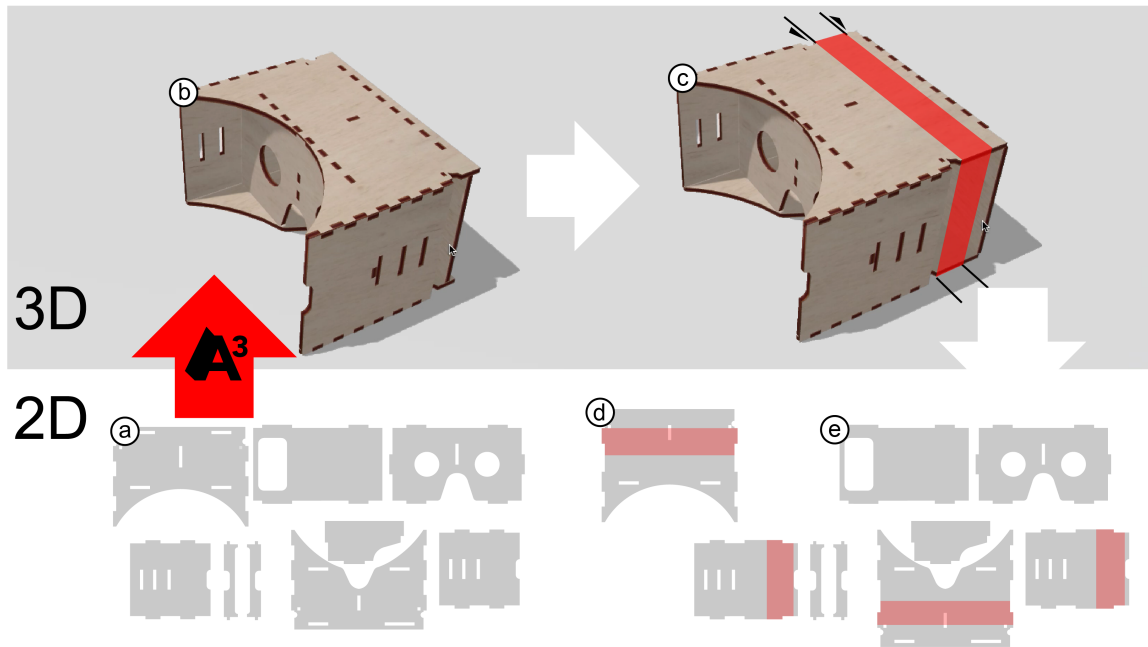


Figure 1: Assembler³ allows users to modify 2D cutting plans by (b) rearranging them into a 3D model, where (c) users can now apply parametric manipulations using existing 3D editor kyub [2], before (d) converting back to 2D for cutting. In our study, this workflow allowed participants to apply parametric modifications 10x faster, more easily (6/7 vs. 2/7), and 26x more reliably, than when employing the traditional workflow of rewriting the 2D cutting plan directly.

ABSTRACT

We demonstrate Assembler³ a software tool that allows users to perform 3D parametric manipulations on 2D laser cutting plans. Assembler³ achieves this by semi-automatically converting 2D laser cutting plans to 3D, where users modify their models using available 3D tools (kyub), before converting them back to 2D. In our user study, this workflow allowed users to modify models 10x faster than using the traditional approach of editing 2D cutting plans

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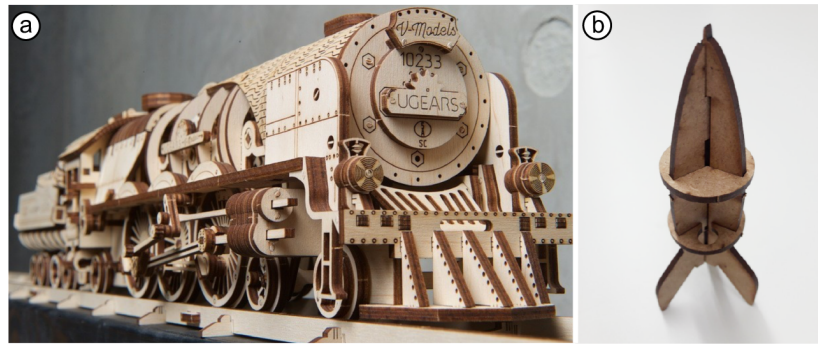


Figure 2: (a) This 3D models designed by a closed design teams (here ugearsmodels.com) consists of 538 parts. (b) The most often fabricated model shared on thingiverse.com (IDs 2440907), in contrast, is of much lower complexity.

directly. Assembler³ converts models to 3D in 5 steps: (1) plate detection, (2) joint detection, (3) material thickness detection, (4) joint matching based on hashed joint "signatures", and (5) interactive reconstruction. In our technical evaluation, Assembler³ was able to reconstruct 100 of 105 models. Once 3D-reconstructed, we expect users to store and share their models *in 3D*, which can simplify collaboration and thereby empower the laser cutting community to create models of higher complexity.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI);

KEYWORDS

laser cutting, personal fabrication, reuse, remixing

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1 INTRODUCTION

While laser cutters are capable of creating complex 3D objects [1], illustrated by the professionally designed model shown in Figure 2a (ugearsmodels.com). Laser-cut models created *by the community*, in contrast, do not reflect this potential: the most often fabricated model on the sharing site thingiverse.com, for example, consists of only 5 parts (Figure 2b).

We feel that this lack of complexity is not coincidental. Communities *are* capable of producing complexity (e.g. the open source code community [3] [5]), but this requires community members to be able to have efficient access to the shared project. In laser cutting, however, 95% of laser cut 3D models are still shared in the form of *2D* cutting plans—a format that makes 3D modifications all but impossible and thus makes it difficult to build on the work of others.

To illustrate this point, we use Figure 1 for visual reference. (a) A user wanting to accommodate the 2D cutting plan of this virtual reality headset for far sightedness might start by (d) extending the top plate of the model but would then also have to realize that (e) three additional plates require modifications as well.

This sounds straightforward, but in order to figure out *which* plates to modify, the user would most likely (b) *mentally* reconstruct the 3D model, (c) *mentally* scale the model in 3D, and (e) *mentally* apply those changes to the 2D cutting plan. Only now could the user realize that the side plates of this particular model have to be scaled, but for example *not* the middle piece. In our user study, it took participants on average 24:45min to complete this particular task, they rated it as "hard" (6/7), and only 2/13 of the resulting cutting plans assembled into functional models.

In this paper, we present Assembler³. Assembler³ is an interactive software tool that implements the "mental" workflow shown in Figure 1 as an actual, *software-based* workflow. (a) Assembler³ allows users to modify *2D* cutting plans by (b) rearranging them into a 3D model, at which (c) users can now apply parametric manipulations using existing 3D editors, before (d) converting back to 2D for cutting. In our study, this workflow allowed participants to apply parametric modifications 10x faster (2:22min on average) than the traditional workflow of rewriting the 2D cutting plan directly. Participants rated the task easy (2/7) and all exported cutting plans assembled into functional models. For the full technical details, we refer to the full paper [4].

2 ASSEMBLER³ WORKFLOW

The main step implemented by Assembler³ is the reconstruction of a 3D model from a 2D cutting plan.

While we present our algorithm in full detail in the section 'The Algorithm of Assembler³', Figure 3 provides a preview. Assembler³ reconstructs a model, such as (a) this plate of the VR headset, by performing the following steps: (b) *plate detection* determines what are plates and what is scrap using the nesting order of paths and assigning plate vs scrap in alternating order. (c) *Joint detection* identifies joint candidates by detecting patterns of left/right turns in the paths. (d) *Material thickness detection* has every joint candidate vote for a material thickness, the dimension most commonly voted for by the joints determines the material thickness. (e) *Joint matching* and,

for fast retrieval, storage in a hash. (f) Interactive reconstruction in a 3D environment (kyub).

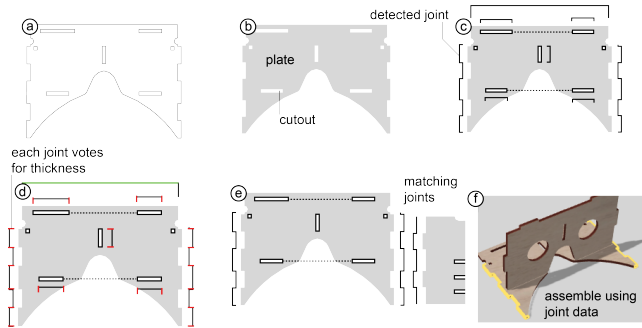


Figure 3: Pipeline of parsing the top plate of the VR headset (in reality it parses the entire SVG).

While we explain each of these steps in detail in the ‘Algorithm’ section, Figure 4 illustrates the last step, i.e. *interactive reconstruction* in (a) the 3D environment. (b) The main interaction is the user attaching two plates to each other using the *assemble tool*. Here the user has selected a plate, which causes it to highlight in yellow. Assembler³ responds by highlighting candidate joints that *could* be paired up with the selected plate. This response is instantaneous, as all matches were precomputed and stored in an efficient-to-retrieve format (joint hash).

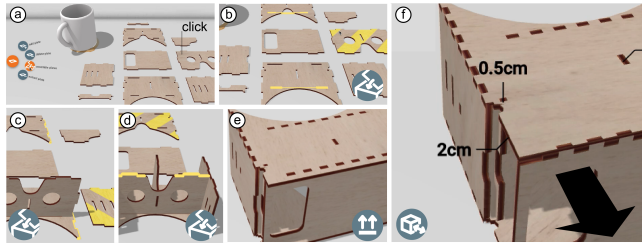


Figure 4: Reconstructing the 3D model of the VR headset. (a) When Assembler³ loads the SVG, all plates are displayed in the 3D modeling environment, here kyub [2]. (b) The user clicked the *assemble tool* on the front piece. Assembler³ responds by highlighting this plate (yellow stripes) and by highlighting joint candidates located on the other plates. (c) Clicking one of the suggested candidates assembles the plate. (d) The user repeats this until the model is assembled. From now on, the user uses standard kyub tools to interact with the model (e) to see the front plate, the user flips the model. (f) Once reconstructed, the user can apply arbitrary parametric changes. Here the user accommodates for far-sightedness by stretching the front plate.

As shown in Figure 4c the user then clicks on a target plate to assemble the selected plate with, Assembler³ responds by attaching the joint candidate to the target plate. The user repeats this step until all plates are assembled into the 3D model.

Once the conversion to 3D is complete, Assembler³ allows users to apply any 3D editing tools to the model. In Figure 1e, the user uses this to accommodate the design for the user’s far-sightedness.

After reconstructing a model once, other users can leverage that effort by now applying *any number* of additional modifications, such as the changes shown in Figure 5. Users may also store and share the model in 3D format, which lowers the bar for others to build on this model, contribute to it, and help it achieve complexity.



Figure 5: When users have converted the VR headset model to 3D *once*, the 3D model allows leveraging existing tools (such as kyub [2]) to perform *any number* of modifications efficiently, such as (a) adjusting the inter-ocular distance, (b) making the headset fit a wider phone by stretching it vertically, (c) or making the headset more comfortable to wear by reducing the weight of the device by decreasing material thickness and removing unnecessary material.

Note how the 3D reconstruction process seamlessly *integrates* into the regular 3D editor environment. Initially, we had expected that 3D reconstruction would have to result in a more wizard/process-funnel like design—a workflow limiting users to back-and-forth navigation along certain process steps. However, once we had fully automated the 2D processing steps, such as material thickness reconstruction, we found the opportunity to implement 3D reconstruction as a *set of tools*, most prominently *assemble tool* and the *floating menu*.

The benefit of this integration is that it invites users to tackle plates in any order, fix mistakes recognized late by disassembling selectively using the “extract plate” tool, use the editor environment to perform any other modeling activities along the way, start parametric manipulations before the import is even complete, or even to reconstruct models only partially for the purpose of remixing rather than full reconstruction.

3 CONCLUSION

In this demonstration, we presented Assembler³ a software tool the primary purpose of which is to allow users to perform parametric manipulations on 2D cutting plans of 3D models in 3D. Assembler³ achieves this by allowing users to convert 2D laser cutting plans to 3D models, modify their models using available 3D tools (kyub), before converting them back to 2D. This makes users 10x faster and results in 26x less errors than the traditional approach of editing 2D cutting plans directly.

Once a laser cut model has been 3D reconstructed using Assembler³ this model will most likely continue its life in this 3D format, making the model easier to process, share, and remix from this point onward. We think that this will help the laser cutting community leave the sharing of 2D cutting plans behind and to transition to a more sharing-friendly and collaboration-friendly

format—which must be a 3D format. We anticipate that this will foster collaboration around models and thus ultimately increase the level of model complexity the laser cutting community will be able to achieve.

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